Experiment: Blocking Hackathon Website Access from CCF Using Extended ACL

Aim

To design and configure a network in Cisco Packet Tracer and use an Extended Access Control List (ACL) to block students in the Central Computing Facility (CCF) from accessing the hackathon website hosted on a server, while allowing other services and network traffic.

Given Information

- College network: 140.80.0.0
- Number of subnets required: 20
- CCF: 4th subnet
- Hackathon server: 7th host address of the 16th subnet
- Website access must be blocked from CCF.
- Other services on the server must remain accessible.

Address Calculation

Since 20 subnets are required, $2^5 = 32$ subnets are available. Therefore, 5 bits are borrowed from the host portion of the Class B network.

Subnet mask

```
255.255.248.0   (/21)
```

Block size = 256 - 248 = 8

CCF - 4th subnet

```
Network address : 140.80.24.0/21
Usable range    : 140.80.24.1 - 140.80.31.254
Router gateway  : 140.80.31.254
```

Hackathon server - 16th subnet

```
Network address : 140.80.120.0/21
7th host address: 140.80.120.7
Server gateway  : 140.80.127.254
```

Network Topology

Devices required: 1 PC (PC0, in the CCF), 1 Server (Server0, the hackathon server), 2 Switches (Switch0 and Switch1, model 2960) and 1 Router (Router0, with at least two Ethernet/Gigabit Ethernet interfaces).

```
PC0 --- Switch0 --- Router0 --- Switch1 --- Server0
(CCF)                                  (Hackathon Server)
```

Cable connections (Copper Straight-Through)

From device	Port	To device	Port
PC0	FastEthernet0	Switch0	FastEthernet0/1
Switch0	FastEthernet0/2	Router0	GigabitEthernet0/0

From device	Port	To device	Port
Router0	GigabitEthernet0/1	Switch1	FastEthernet0/1
Switch1	FastEthernet0/2	Server0	FastEthernet0

Procedure (Cisco Packet Tracer)

Step 1: Build the topology

1. Open Cisco Packet Tracer and start with a new, empty workspace.
2. From End Devices, drag one PC (rename to PC0) and one Server (rename to Server0) into the workspace.
3. From Network Devices, add two 2960 switches (rename to Switch0 and Switch1) and one router (rename to Router0).
4. Using the Connections icon, join the devices with Copper Straight-Through cables exactly as given in the cable connection table above.

Step 2: Configure Router0 interfaces

Open Router0 -> CLI and enter only the commands (not the prompt):

```
enable
configure terminal
interface GigabitEthernet0/0
ip address 140.80.31.254 255.255.248.0
no shutdown
exit
interface GigabitEthernet0/1
ip address 140.80.127.254 255.255.248.0
no shutdown
exit
end
show ip interface brief
```

Expected: both GigabitEthernet0/0 (140.80.31.254) and GigabitEthernet0/1 (140.80.127.254) show status **up** and protocol **up**. If an interface is down, check the cable connection.

Step 3: Configure PC0 and Server0

Open Desktop -> IP Configuration on each device and enter:

Device	IP Address	Subnet Mask	Default Gateway
PC0	140.80.24.1	255.255.248.0	140.80.31.254
Server0	140.80.120.7	255.255.248.0	140.80.127.254

Step 4: Enable the website service

On Server0, open Services -> HTTP and make sure **HTTP: On** (and HTTPS: On, if available). This allows the ACL's effect on web traffic to be tested.

Step 5: Create the Extended ACL

On Router0 CLI, enter global configuration mode and create ACL 100:

```
enable
configure terminal
access-list 100 deny tcp 140.80.24.0 0.0.7.255 host 140.80.120.7 eq 80
```

```
access-list 100 deny tcp 140.80.24.0 0.0.7.255 host 140.80.120.7 eq 443
access-list 100 permit ip any any
```

The first two lines block HTTP (port 80) and HTTPS (port 443) from the CCF subnet to the server. The last line permits all other traffic, since every ACL ends with an implicit "deny all".

Step 6: Apply the ACL

The CCF is connected to GigabitEthernet0/0, so the ACL is applied inbound on that interface:

```
interface GigabitEthernet0/0
ip access-group 100 in
end
```

Step 7: Save the configuration

```
enable
copy running-config startup-config
```

Press Enter at the prompt *Destination filename [startup-config]?* and wait for *Building configuration... [OK]*.

Verification and Testing

Verify the ACL

```
show access-lists 100
```

Expected output:

```
Extended IP access list 100
    deny tcp 140.80.24.0 0.0.7.255 host 140.80.120.7 eq www (62 match(es))
    deny tcp 140.80.24.0 0.0.7.255 host 140.80.120.7 eq 443
    permit ip any any (24 match(es))
```

The match counters confirm that traffic is actually passing through the ACL.

Test 1 - Ping from PC0 to the router

PC0 -> Desktop -> Command Prompt:

```
ping 140.80.31.254
```

Expected: *Reply from 140.80.31.254*

Test 2 - Ping from PC0 to the server

```
ping 140.80.120.7
```

Expected: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss). This confirms that ping (ICMP) is allowed.

Test 3 - Open the website from PC0

PC0 -> Desktop -> Web Browser, enter `http://140.80.120.7` and click Go. The webpage should **not** load, which is the desired result because HTTP traffic is denied by ACL 100.

Why ping works but the website does not

- Ping uses ICMP. The ACL does not deny ICMP, so ping works (permitted by `permit ip any any`).
- The website uses TCP port 80 (HTTP) and TCP port 443 (HTTPS). The ACL specifically denies these connections from the CCF network to the server.

Traffic from CCF to Server	Result
Ping / ICMP	ALLOWED
HTTP / TCP 80	BLOCKED
HTTPS / TCP 443	BLOCKED

This satisfies the requirement to block only website access without denying other services.

Conclusion

The experiment successfully demonstrated how an Extended Access Control List can be used to selectively control network access. By specifying the source subnet, destination server, protocol, and port number, only HTTP and HTTPS access from the CCF to the hackathon server was blocked. Other communication with the server remained available, satisfying the requirement that the server's other services should not be denied. The experiment achieved CO3 by applying networking concepts to configure and test access control on a network.

Result

The network was successfully configured in Cisco Packet Tracer using an Extended ACL. The CCF subnet 140.80.24.0/21 was prevented from accessing the hackathon server's web services at 140.80.120.7 on TCP ports 80 and 443. Other IP traffic, including ICMP ping, remained permitted. The ACL counters confirmed that the HTTP deny rule was being matched, while the successful ping confirmed that other traffic was not being blocked.